



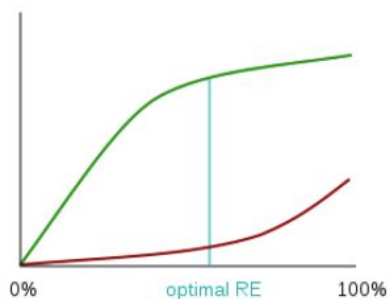
**SCIENCE OLYMPIAD**  
— AT THE —  
**UNIVERSITY OF FLORIDA**

Northern Regional: January 19<sup>th</sup>, 2019

Water Quality C  
Picture Packet

**DO NOT WRITE ON THIS**

### Station 3 (20 Points)



1. Refer to the above graph.
  - a. What type of reproduction is modeled by the graph? (1)
  - b. Label the x-axes and each of the lines. What determines the “optimal RE”? (3)
  - c. (True/False) This type of reproduction is consistent with that of a salmon. (1)
  - d. Is this type of reproduction consistent with r or K selection? (1)

| Community | Species 1 | Species 2 | Species 3 | Species 4 | Species 5 |
|-----------|-----------|-----------|-----------|-----------|-----------|
| <i>x</i>  | 3         | 4         | 2         | 0         | 0         |
| <i>y</i>  | 3         | 3         | 0         | 4         | 1         |

2. Refer to the above chart. A researcher visits two communities and collects data regarding the populations of five different species in the two communities.
  - a. What is Sorensen’s coefficient of similarity for these two communities? (2)
  - b. What are the maximum and minimum values of Sorensen’s coefficient? What do they signify? (2)
  - c. Calculate Simpson’s index of diversity for both communities. Is it appropriate to compare the two values, and why? (3)



## Station 7 (31 Points)

A



B



C



D



E



F



1. Refer to organism A.
  - a. What is the common name of this organism? (1)
  - b. What is the lifespan of these adult organisms? (3)
  - c. The larvae of this organism are some of the (largest/smallest). (1)
2. Refer to organism B.
  - a. Identify the phylum this organism belongs to. (1)
  - b. What is the feeding strategy of this organism? (2)
  - c. What is the reproductive strategy of this organism? (2)
3. Refer to organism C.
  - a. What organism is this? (1)
  - b. What are the nicknames of the larvae of this organism? (2)
  - c. What are differences between the males and females in this species? (2)